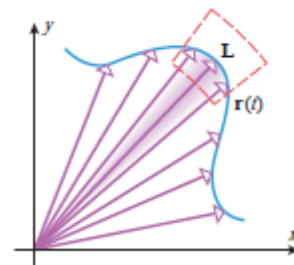


2.2 Calculus of Vector-Valued Functions

Recall: A vector-valued function $\mathbf{r}(t)$ maps real numbers to n -dimensional vectors $\mathbf{r}: \mathbb{R} \rightarrow \mathbb{R}^n$.

It is important to visualize how $\mathbf{r}(t)$ plots points via vectors.

We begin this section by looking at the limit of a vector-valued function.



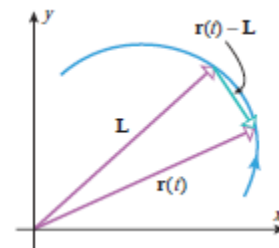
Definition: Let $\mathbf{r}(t)$ be a vector-valued function that is defined for all t in some open interval containing the number a (it need not be defined at a). We write that

$$\lim_{t \rightarrow a} \mathbf{r}(t) = \mathbf{L}$$

if and only if

$$\lim_{t \rightarrow a} \|\mathbf{r}(t) - \mathbf{L}\| = 0$$

Note: It is important to see that the limit of $\mathbf{r}(t)$ is a vector $\mathbf{L}$. $\mathbf{L}$ is often referred to as a limiting vector. Therefore, we don't imagine we following points on the curve when we evaluate the limit of $\mathbf{r}(t)$, but rather we are following the sequence of vectors since $\mathbf{r}(t)$ is a function that produces vectors.



The above definition suggests the following theorem.

Theorem: If $\mathbf{r}(t) = \langle f(t), g(t), h(t) \rangle$, then we define $\lim_{t \rightarrow a} \mathbf{r}(t)$ as follows:

$$\lim_{t \rightarrow a} \mathbf{r}(t) = \left\langle \lim_{t \rightarrow a} f(t), \lim_{t \rightarrow a} g(t), \lim_{t \rightarrow a} h(t) \right\rangle.$$

Proof: Omitted, but it follows from the above definition.

Definition: A vector function $\mathbf{r}(t)$ is continuous at a if $\lim_{t \rightarrow a} \mathbf{r}(t) = \mathbf{r}(a)$.

Theorem: A vector function $\mathbf{r}$ is continuous at a if and only if its component functions are continuous at a .

Proof: This theorem follows immediately from how continuity is defined for vector functions and how continuity is defined in calculus one for functions of the form $f: \mathbb{R} \rightarrow \mathbb{R}$.

Example 1: Consider the vector function $\mathbf{r}(t) = \left(\frac{t^2 - t}{t - 1} \right) \mathbf{i} + (\sqrt{t + 8}) \mathbf{j} + \left(\frac{\sin(\pi t)}{\ln(t)} \right) \mathbf{k}$.

a) Find $\lim_{t \rightarrow 1} \mathbf{r}(t)$

We use definition above the properties of limits from calculus one and l'Hopital's rule to discover:

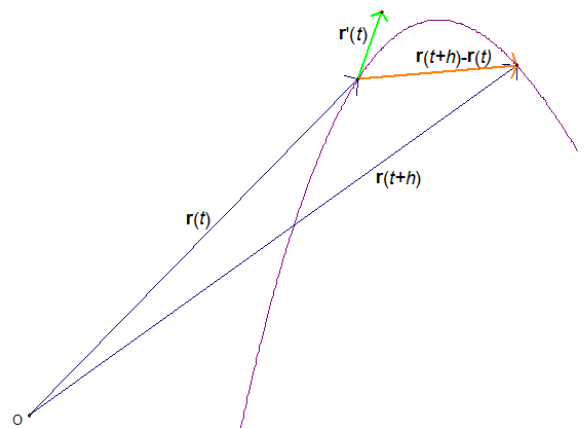
$$\begin{aligned} \lim_{t \rightarrow 1} \mathbf{r}(t) &= \left(\lim_{t \rightarrow 1} \frac{t^2 - t}{t - 1} \right) \mathbf{i} + \left(\lim_{t \rightarrow 1} \sqrt{t + 8} \right) \mathbf{j} + \left(\lim_{t \rightarrow 1} \frac{\sin(\pi t)}{\ln(t)} \right) \mathbf{k} \stackrel{H}{=} \left(\lim_{t \rightarrow 1} t \right) \mathbf{i} + (\sqrt{9}) \mathbf{j} + \left(\lim_{t \rightarrow 1} \frac{\pi \cos(\pi t)}{(1/t)} \right) \mathbf{k} \\ &= (1) \mathbf{i} + (3) \mathbf{j} + (-\pi) \mathbf{k} = \mathbf{i} + 3\mathbf{j} - \pi \mathbf{k}. \end{aligned}$$

b) Is the function $\mathbf{r}$ continuous at 1? Give a brief explanation of your answer.

We note that $\mathbf{r}(1)$ is not defined. Thus, $\lim_{t \rightarrow 1} \mathbf{r}(t) \neq \mathbf{r}(1)$, and $\mathbf{r}$ is not continuous at 1 by definition.

Definition: The derivative of a vector function $\mathbf{r}$ is defined by: $\frac{d\mathbf{r}}{dt} = \mathbf{r}'(t) = \lim_{h \rightarrow 0} \frac{\mathbf{r}(t+h) - \mathbf{r}(t)}{h}$.

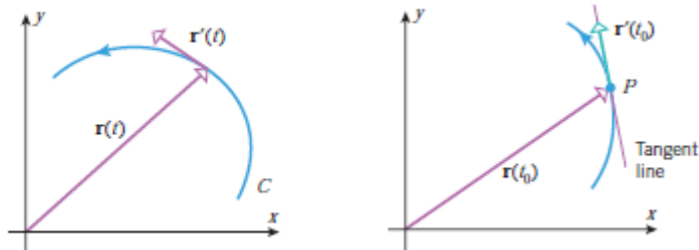
Intuitively, one can imagine that as h approaches 0 the vector in orange will approach the vector in green (which is tangent to the curve at the terminal point of $\mathbf{r}(t)$). This leads naturally to the following definitions.



Definition: Suppose that $\mathbf{r}'(t) \neq \mathbf{0}$. The vector $\mathbf{r}'(t)$ is called the tangent vector to the curve, C , defined by $\mathbf{r}$ at the terminal point, P , of the position vector $\mathbf{r}(t)$.

The tangent line to C at P is defined to be the line through P parallel to $\mathbf{r}'(t)$.

Also, $\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{\|\mathbf{r}'(t)\|}$ is called the unit tangent vector.



The next theorem gives us a familiar way to calculate $\mathbf{r}'(t)$.

Theorem: If $\mathbf{r}(t) = \langle f(t), g(t), h(t) \rangle$ and the component functions of $\mathbf{r}$ are differentiable then,
 $\mathbf{r}'(t) = \langle f'(t), g'(t), h'(t) \rangle$.

Proof:

Using the definition of $\mathbf{r}'(t)$ we calculate:

$$\begin{aligned} \mathbf{r}'(t) &= \lim_{k \rightarrow 0} \frac{\mathbf{r}(t+k) - \mathbf{r}(t)}{k} = \lim_{k \rightarrow 0} \frac{1}{k} (\langle f(t+k), g(t+k), h(t+k) \rangle - \langle f(t), g(t), h(t) \rangle) \\ &= \lim_{k \rightarrow 0} \frac{1}{k} (\langle f(t+k) - f(t), g(t+k) - g(t), h(t+k) - h(t) \rangle) \\ &= \lim_{k \rightarrow 0} \left\langle \frac{f(t+k) - f(t)}{k}, \frac{g(t+k) - g(t)}{k}, \frac{h(t+k) - h(t)}{k} \right\rangle \\ &= \left\langle \lim_{k \rightarrow 0} \frac{f(t+k) - f(t)}{k}, \lim_{k \rightarrow 0} \frac{g(t+k) - g(t)}{k}, \lim_{k \rightarrow 0} \frac{h(t+k) - h(t)}{k} \right\rangle. \end{aligned}$$

Since the component functions of $\mathbf{r}$ are differentiable we know

$$\left\langle \lim_{k \rightarrow 0} \frac{f(t+k) - f(t)}{k}, \lim_{k \rightarrow 0} \frac{g(t+k) - g(t)}{k}, \lim_{k \rightarrow 0} \frac{h(t+k) - h(t)}{k} \right\rangle = \langle f'(t), g'(t), h'(t) \rangle.$$

■

Note: There is nothing special in the above proof that requires the vector function to have 3 components. The same proof works for a vector function of the form $\mathbf{r}: \mathbb{R} \rightarrow \mathbb{R}^n$.

Example 2: Consider $\mathbf{r}(t) = \langle 2 \cos(t), \sin(t), t \rangle$

a) Find $\mathbf{r}'(t)$

We calculate: $\mathbf{r}'(t) = \langle -2 \sin(t), \cos(t), 1 \rangle$.

b) Find $\mathbf{T}(t)$

We calculate

$$\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{\|\mathbf{r}'(t)\|} = \frac{1}{\sqrt{4 \sin^2(t) + \cos^2(t) + 1}} \langle -2 \sin(t), \cos(t), 1 \rangle = \frac{1}{\sqrt{3 \sin^2(t) + 2}} \langle -2 \sin(t), \cos(t), 1 \rangle \text{ provided } \mathbf{r}'(t) \neq \mathbf{0}.$$

c) Find parametric equations for the tangent line to the curve determined by $\mathbf{r}$ at the point $\left(0, 1, \frac{\pi}{2}\right)$.

A direction vector parallel to the desired line is $\mathbf{r}'\left(\frac{\pi}{2}\right) = \langle -2, 0, 1 \rangle$. So, a vector equation for the line is

$$\langle x, y, z \rangle = \left\langle 0, 1, \frac{\pi}{2} \right\rangle + t \langle -2, 0, 1 \rangle, \text{ and parametric equations for the line are } x = -2t, y = 1, z = \frac{\pi}{2} + t.$$

Note: You can use the graphing software on Blackboard to help visualize what we have found in this example.

Theorem: Suppose $\mathbf{u}: \mathbb{R} \rightarrow \mathbb{R}^3$ and $\mathbf{v}: \mathbb{R} \rightarrow \mathbb{R}^3$ are differentiable vector functions, c is a scalar, and $f: \mathbb{R} \rightarrow \mathbb{R}$ is a differentiable function. Then,

1. $\frac{d}{dt}[\mathbf{u}(t) + \mathbf{v}(t)] = \mathbf{u}'(t) + \mathbf{v}'(t)$
2. $\frac{d}{dt}[c\mathbf{u}(t)] = c\mathbf{u}'(t)$
3. $\frac{d}{dt}[f(t)\mathbf{u}(t)] = f'(t)\mathbf{u}(t) + f(t)\mathbf{u}'(t)$
4. $\frac{d}{dt}[\mathbf{u}(t) \cdot \mathbf{v}(t)] = \mathbf{u}'(t) \cdot \mathbf{v}(t) + \mathbf{u}(t) \cdot \mathbf{v}'(t)$
5. $\frac{d}{dt}[\mathbf{u}(t) \times \mathbf{v}(t)] = \mathbf{u}'(t) \times \mathbf{v}(t) + \mathbf{u}(t) \times \mathbf{v}'(t)$
6. $\frac{d}{dt}[\mathbf{u}(f(t))] = f'(t)\mathbf{u}'(f(t))$

Proof: Many of these are left as practice in the exercises for this section.

Note: With the exception of property 5 all of the above properties hold when $\mathbf{u}$ and $\mathbf{v}$ have two or more component functions.

Theorem: If $\mathbf{r}(t)$ is a differentiable vector-valued function in 2-space or 3-space and $\|\mathbf{r}(t)\|$ is constant for all t , then

$$\mathbf{r}(t) \cdot \mathbf{r}'(t) = 0$$

that is, $\mathbf{r}(t)$ and $\mathbf{r}'(t)$ are orthogonal vectors for all t .

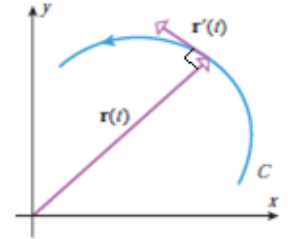
Proof:

By property 4 in the theorem above we know that:

$$\begin{aligned} \frac{d}{dt} [\mathbf{r}(t) \cdot \mathbf{r}(t)] &= \mathbf{r}'(t) \cdot \mathbf{r}(t) + \mathbf{r}(t) \cdot \mathbf{r}'(t) \\ \Rightarrow \frac{d}{dt} [\|\mathbf{r}(t)\|^2] &= 2[\mathbf{r}'(t) \cdot \mathbf{r}(t)] \end{aligned}$$

However, since $\|\mathbf{r}(t)\|^2$ is a constant its derivative is 0 and therefore,

$$\begin{aligned} 0 &= 2[\mathbf{r}'(t) \cdot \mathbf{r}(t)] \\ \Rightarrow 0 &= \mathbf{r}'(t) \cdot \mathbf{r}(t) \end{aligned}$$



Definition: The definite integral of a continuous vector function $\mathbf{r}(t) = \langle f(t), g(t), h(t) \rangle$ is

$$\int_a^b \mathbf{r}(t) dt = \left\langle \int_a^b f(t) dt, \int_a^b g(t) dt, \int_a^b h(t) dt \right\rangle.$$

Note: For now we will hold off on a physical interpretation of the definite integral (we will see a nice application for it in 2.6). The indefinite integral is defined similarly.

Example 3: Suppose $\mathbf{r}(t) = (\cos(t))\mathbf{i} + \left(\frac{1}{1+t^2}\right)\mathbf{j} + (\sin^2(t))\mathbf{k}$. Find $\int_0^{\pi/4} \mathbf{r}(t) dt$ (exactly).

We calculate

$$\begin{aligned} \int_0^{\pi/4} \mathbf{r}(t) dt &= \left(\int_0^{\pi/4} \cos(t) dt \right) \mathbf{i} + \left(\int_0^{\pi/4} \frac{1}{1+t^2} dt \right) \mathbf{j} + \left(\int_0^{\pi/4} \sin^2(t) dt \right) \mathbf{k} \\ &= \left(\sin(t) \Big|_0^{\pi/4} \right) \mathbf{i} + \left(\tan^{-1}(t) \Big|_0^{\pi/4} \right) \mathbf{j} + \left(\frac{1}{2} \int_0^{\pi/4} (1 - \cos(2t)) dt \right) \mathbf{k} = \frac{\sqrt{2}}{2} \mathbf{i} + \tan^{-1}\left(\frac{\pi}{4}\right) \mathbf{j} + \left(\frac{1}{2} \left(t - \frac{1}{2} \sin(2t) \right) \Big|_0^{\pi/4} \right) \mathbf{k} \\ &= \frac{\sqrt{2}}{2} \mathbf{i} + \tan^{-1}\left(\frac{\pi}{4}\right) \mathbf{j} + \left(\frac{\pi}{8} - \frac{1}{4} \right) \mathbf{k} \\ \text{OR} \\ &= \left\langle \frac{\sqrt{2}}{2}, \tan^{-1}\left(\frac{\pi}{4}\right), \frac{\pi}{8} - \frac{1}{4} \right\rangle \end{aligned}$$

Note: Again, it is important to see that our result is a vector.

Example 4: Suppose $\mathbf{r}(t) = (\cos(t))\mathbf{i} + \left(\frac{1}{1+t^2}\right)\mathbf{j} + (\sin^2(t))\mathbf{k}$. Find $\int \mathbf{r}(t) dt$.

We already know the antiderivatives of each component from Example 3. So we have that

$$\begin{aligned}\int \mathbf{r}(t) dt &= [\sin(t) + C_1]\mathbf{i} + [\tan^{-1}(t) + C_2]\mathbf{j} + \left(\frac{1}{2}\left(t - \frac{1}{2}\sin(2t)\right) + C_3\right)\mathbf{k} \\ &= \left\langle \sin(t) + C_1, \tan^{-1}(t) + C_2, \frac{1}{2}t - \frac{1}{4}\sin(2t) + C_3 \right\rangle\end{aligned}$$

OR

$$\begin{aligned}\int \mathbf{r}(t) dt &= \left\langle \sin(t), \tan^{-1}(t), \frac{1}{2}t - \frac{1}{4}\sin(2t) \right\rangle + \langle C_1, C_2, C_3 \rangle \\ &= \left\langle \sin(t), \tan^{-1}(t), \frac{1}{2}t - \frac{1}{4}\sin(2t) \right\rangle + \mathbf{C}\end{aligned}$$

Where $\mathbf{C} = \langle C_1, C_2, C_3 \rangle$ is a constant vector.

Note: If we had been given some initial condition regarding the vector-function $\int \mathbf{r}(t) dt$, then we could have solved for the values of all the unknown constants. This is just like initial condition questions from Calculus I.